Measurement of propagation speed, absorption and depth determination in various materials using ultrasound

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In this experiment, the speed of sound was measured in the different materials polyacrylic, water, aluminum, and copper. A GS200 ultrasound echoscope was used to generate a sound wave with a fixed frequency. A minimal dispersion was observed when comparing the speeds at different frequencies. A B-scan was performed on a test block and, using the same test pattern, it was confirmed that lengths could be accurately determined using ultrasound up to the resolution limit through time of flight measurements, which were compared with caliper measurements. Spectral methods based on Fourier transformation were used to determine the depth of a polyacrylic plate.

1 Introduction

Ultrasound refers to mechanical waves with very high frequencies that cannot be heard by humans, as they exceed 20 kHz. In contrast to light waves, ultrasound requires a physical medium in which atoms or molecules are connected and able to vibrate in order for the waves to move through it. Depending on the material, these waves may propagate in various transverse or longitudinal modes. In crystals with certain symmetry directions, they generally show unique dispersion behaviors and characteristic sound speeds. Ultrasound is widely applied in fields such as medical imaging and material evaluation, especially for nondestructive inspection. The purpose of this experiment is to investigate the essential concepts behind material analysis and ultrasound based imaging.

2 Basic functions of an echoscope

An echoscope is an ultrasound measuring device that connects a probe to a computer or oscilloscope, and a piezoelectric crystal is typically used as the transducer. The probe emits ultrasonic waves when an electrical signal of the appropriate frequency is applied by the echoscope, causing the piezoelectric crystal to vibrate. At the same time, the probe also acts as a detector: returning ultrasonic waves are converted back into electrical signals, which the echoscope amplifies, processes, and forwards to the computer.
The echoscope can operate in two primary modes. In reflection mode, a single probe serves simultaneously as transmitter and receiver, capturing echoes created at material boundaries. In transmission mode, two separate probes are used, placed on opposite sides of the sample so that one emits and the other receives the transmitted ultrasound.
Various methods are available to analyze the signals recorded by the echoscope. This experiment focuses on the two most common methods. The first is the A-scan (amplitude scan), where the probe remains fixed and the amplitude of the received signal is displayed as a function of time after a pulse has been sent. The second is the B-scan (brightness scan), in which the probe is moved across the sample, producing a two dimensional representation of the scanned area. In this display, brightness levels correspond to signal amplitudes, giving the B-scan its characteristic appearance.

3 Measuring the velocity of sound in different materials

In this part of the experiment, the speed of ultrasound is investigated in different materials. For this purpose, the GS200 ultrasound echoscope with frequencies of 1MHz, 2MHz and 4MHz is used.

3.1 Motivation

The precise determination of the speed of sound in materials is crucial for applications such as non destructive testing and ultrasonic diagnostics. Since the speed of sound is directly influenced by materialdependent properties, its measurement provides important information about the structure and condition of a material. Using an ultrasonic echo device, these speeds can be determined accurately and non invasively. The data obtained enables both comparison with theoretical models and improved characterization of the materials under investigation.

3.2 Speed of sound in polyacryl

An ultrasound pulse was sent into polyacrylic cylinders of different sizes. The echoscope can be used to measure the travel time Δt of the reflected sound pulses, and the speed of sound v_s in polyacrylic can be calculated using equation 1.

$$v_s = \frac{2l}{\Delta t} \tag{1}$$

To determine the distance l traveled by the piston, the cylinders were measured with a caliper. The dimensions can be found in table 1.

Cylinder	Length [mm]	Diameter [mm]
1	41,30	40,73
2	80,81	40,40
3	120,76	40,74

Table 1: Dimensions of the cylinders measured with a caliper

The caliper error is $\sigma_c = 0.005mm$ and thus also the diameter and length error. The time difference error is $\sigma_t = 1\mu s$. Using the time differences determined in table 2, the cylinder lengths and equation 1, the speed of sound can be determined as a function of frequency. The values are in table 4.

Cylinder	Δt_{1MHz} [μs]	Δt_{2MHz} [μs]
1	31	31,5
2	60	60
3	92	90

Table 2: Determined time difference between sending the signal and detecting the reflection at different frequencies

Cylinder	Δt_{4MHz} [μs]
1	30,5
2	60
3	-

Table 3: Determined time difference between sending the signal and detecting the reflection at $f = 4MHz$

Frequency [MHz]	Velocity [$\frac{km}{s}$]
1	$2,646 \pm 0,012$
2	$2,681 \pm 0,012$
4	$2,697 \pm 0,020$

Table 4: Calculated sound velocity v_s in a polyacrylic cylinder for various high frequency sound waves

To find a linear dispersion, the velocity values were plotted as a function of frequency in figure 1 and a regression line was determined using linear regression. Equation 2 describes the speed of sound in dependence on frequency in polyacryl. The errors of the velocity are calculated with gaussian error calculation. Table 3 lacks one value because no reflection of the signal could be found. This is probably because of the damping of this frequency, which lowers the amplitude of the signal at this cylinder length, so it is indistinguishable from the noise.

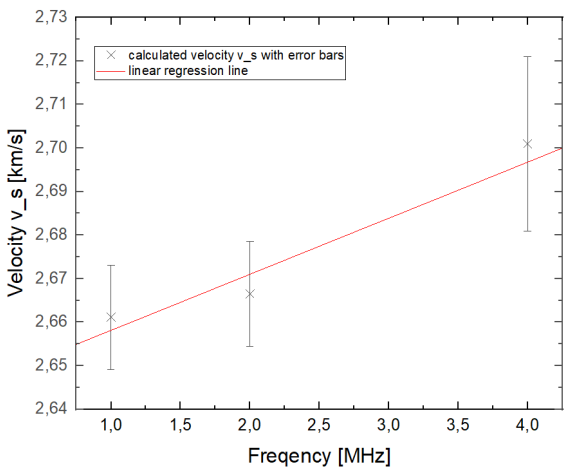

Figure 1: Determination of a linear dispersion between frequency and sound velocity in polyacrylic

$$v_s = 0,0129km * f + 2,6453\frac{km}{s} \tag{2}$$

If equation 2 has the form $v_s = a * f + b$, then one must take into account the errors $\sigma_a = 0,0039km$ and $\sigma_b = 0,0082\frac{km}{s}$.

3.3 Speed of sound in water

To determine the velocity of ultrasound in water, water got filled in a plastic box and a reflector was put inside. The distance between the probe and the reflector is variable. Only the frequency of $4MHz$ was used, because in 2.1 it was observed that the dispersion has only a minimal effect on the propagation speed.

Distance [mm]	Δt [μs]	v_w [$\frac{km}{s}$]
29,95	42,5	$1,409 \pm 0,033$
40,08	56,0	$1,431 \pm 0,023$
60,00	83,5	$1,437 \pm 0,017$
80,07	110,1	$1,454 \pm 0,013$
107,51	147,3	$1,4597 \pm 0,0099$

Table 5: Measured time difference between sending a sound wave of $f = 4MHz$ and receiving the reflection in water. From the time difference and the distance traveled, the speed of sound in water was determined using (1).

With equation 1 and the values from table 5, the ultrasound velocity v_w can be calculated in water. The errors for the distance and the time difference are the same as in 2.1. The results are also in table 5. The weighted average was calculated using the errors and the best values. The velocity of ultrasound in water is $v_w = (1,4507 \pm 0,0068)\frac{km}{s}$. The value in the literature is $v_{w_{Lit}} = 1,461\frac{km}{s}$ and is therefore not compatible with the determined value. The main reason why the values do not match is due to the composition of the water. To determine the literature value, pure water was used, whereas in this experiment, highly calcareous water was used.

3.4 Speed of sound in cooper and aluminum

Copper and aluminum are crystalline materials with a periodic atomic structure. Because the bonding strength between atoms depends on the crystallographic direction, sound waves propagate at different velocities, depending on their mode of vibration. Accurately determining these velocities therefore requires a more sophisticated measurement technique. The measurement setup is similar to the setup in 2.2 and is shown in figure 7.

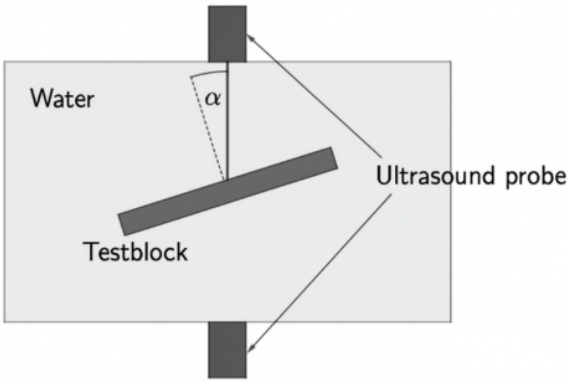

Figure 2: Experimental setup for determining the speed of sound in solids made of copper and aluminum

To measure the velocity of sound in the solid, a block of the material under investigation is submerged in water at different angles, and the amplitudes of an A-scan are recorded in transmission mode. At the boundary between the materials, the angles of the incident wave (α)

and the refrected wave (β) obey Snell's law.

$$\frac{sin(\beta)}{sin(\alpha)} = \frac{v_s}{v_w} \tag{3}$$

v_w is the known speed of sound in water and v_s is the known speed of sound in the solid. When performing an A-scan, three peaks are expected, each corresponding to a different wave mode. The peak with the shortest runtime is associated with the longitudinal mode, while the two peaks with longer runtimes correspond to transverse modes. Additionally, it is known that total reflection occurs at $\beta = \frac{\pi}{2}$ for both longitudinal and transverse waves. Consequently, by determining the angle at which the signal of the corresponding mode disappears, $\beta = \frac{\pi}{2}$ can be approximated and v_s calculated. Furthermore, for transverse waves, transmission reaches a maximum when $\beta = 45$. Following the same approach, another value for the transverse mode at the maximum amplitude of the transverse wave peak can be determined.

Medium	Mode	$\beta[°]$	v$[\frac{km}{s}]$
Al	transverse	90	$2,90 \pm 0,11$
Al	transverse	90	$3,31 \pm 0,27$
Al	longitudinal	45	$6,56 \pm 0,59$
Cu	transverse	90	$2,052 \pm 0,093$
Cu	longitudinal	90	$4,69 \pm 0,23$

Table 6: Speed of sound of transverse and longitudinal waves in solids made of copper and aluminum

The experimental results are consistent with theoretical expectations, as the measured values for both longitudinal and transverse wave modes in copper and aluminum fall within the expected theoretical ranges, considering the given margins of error.

4 Absorption of ultrasound in polyacryl

In this part of the experiment the absorption of sound waves in polyacryl is examined. The same frequency and cylinders as in 3.1 are used, and a decrease in the amplitude of the reflected signal is observed. For the frequencies of 1 MHz and 2 MHz three values are determined for different cylinder lengths, this is not the case for 4 MHz, because of this it is only possible to construct the exponential fit for 1 MHz and 2 MHz. The coefficient λ is determined from the exponential fit.

Frequency	λ [cm^{-1}]
1 MHz	-0.460 ± 0.075
2 MHz	-0.668 ± 0.027

Table 7: Absorption of ultrasound in polyacryl for different frequencies

5 Resolution by ultrasonic waves

The wave pulses emitted by the ultrasound device have a finite temporal extent and therefore do not correspond to idealized δ-functions. As a result, reflections from closely spaced structures can no longer be clearly separated, as their corresponding signals overlap and are thus not distinguishable.

To determine the axial resolution of an ultrasound measurement, the full width at half maximum (FWHM) of the resulting signal curve is used, as this remains constant even when the gain changes. Two signals are considered separable from each other if their distance is at least equal to this FWHM.

The full width at half maximum (FWHM) was measured for different sound frequencies in the third bore hole of the test block shown in figure 3. The corresponding measurements are listed in table 8.

Frequency [MHz]	FWHM [μs]
1	$1,53 \pm 0,15$
2	$0,99 \pm 0,15$
4	$0,62 \pm 0,15$

Table 8: Determination of the resolving power of ultrasound waves at different frequencies using the half width on a medium sized cylinder.

With increasing frequency, resolution quality improves.

6 Non destructive measurement of a body

Ultrasound makes it possible to look inside the body without destroying it. To learn this technique, a polyacryl block with eleven holes was used. The depth of every hole and the size of the whole block were measured. Ultrasound and a caliper were used for comparison.

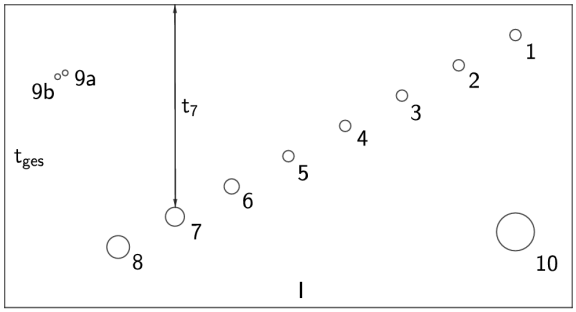

Figure 3: Schematic structure of the measured test block made of polyacrylic

6.1 B-Picture

A B-Scan of the test block was performed. The result is shown in figure 4. The scan shows red spots, where the material reflects the sound waves. In this case, these are bores in the test block, but in industrial applications they can be material defects or filling levels of liquids, while in medicine, B-scans are used to identify different structures inside a living body.

6.2 Length measurements

In section 3.2 the speed of sound in polyacryl was measured. With this velocity and the time difference between sending and receiving the sound wave, the depths d_i of the drilling were calculated using equation 1. With different frequencies, the difference between holes $9a$ and $9b$ was measured. The results are in table 9. To measure the time difference of the signal, an A-Scan and a 2MHz frequency signal were used. Due to the precision of the reading, the time error was estimated at $\sigma_t = \pm 1\mu s$. With these settings, most of the measured values are consistent with the values measured with the caliper.

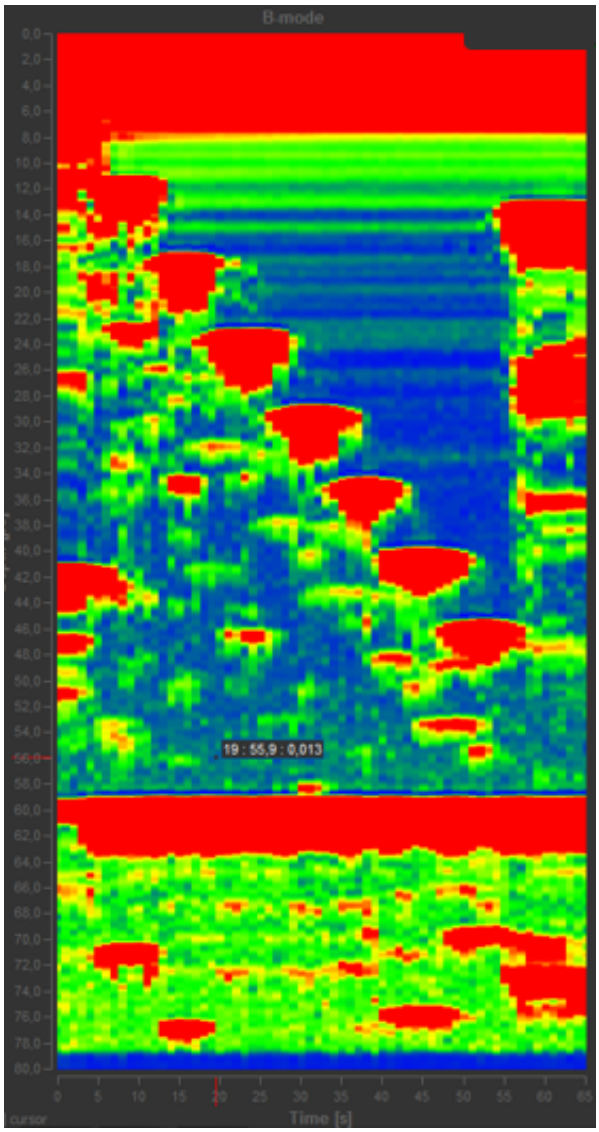

Figure 4: B-Image of the measured test block made of polyacrylic. The test block was schematically shown in Figure (3).

Bore	$d_{caliper}[mm]$	$d_{ultrasound}[mm]$
d_{tot}	$80,26 \pm 0,005$	$80,0 \pm 1,4$
d_1	$7,32 \pm 0,005$	$6,8 \pm 1,3$
d_2	$15,21 \pm 0,005$	$14,7 \pm 1,3$
d_3	$23,55 \pm 0,005$	$22,7 \pm 1,3$
d_4	$31,12 \pm 0,005$	$30,8 \pm 1,3$
d_5	$39,11 \pm 0,005$	$38,1 \pm 1,4$
d_6	$46,76 \pm 0,005$	$46,1 \pm 1,4$
d_7	$54,2 \pm 0,005$	$53,4 \pm 1,4$
d_8	$61,93 \pm 0,005$	$60,2 \pm 1,4$
d_{9a}	$17,65 \pm 0,005$	$17,4 \pm 1,3$
d_{9b}	$19,28 \pm 0,005$	$19,1 \pm 1,3$
d_{10}	$55,69 \pm 0,005$	$54,8 \pm 1,4$
Δd_9	$1,6300 \pm 0,0071$	$1,9845 \pm 0,0019$
Δd_9	$1,6300 \pm 0,0071$	$1,6086 \pm 0,0019$
Δd_9	$1,6300 \pm 0,0071$	$1,6856 \pm 0,0019$

Table 9: Length measurements on a testblock of polyacryl with a caliper and ultrasound $f = 2MHz$. The difference between bore 9a and 9b are meassured with $f = 1MHz$ (top), $f = 2MHz$ (middle) and $f = 4MHz$ (bottom)

6.3 Spectral Methods

Other methods for length measurement with ultra sonic sound waves are spectral methods such as the spectral method determined Fast Fourier Transformation (FFT) spectrum F(g)(t) of a signal g(t), in which the harmonic frequencies of the reflections are clearly recognizable, as well as the cepstrum $C(t) := F^{-1}1(log(F(g))(t)$, in which the relevant period in the frequency can be determined. Through cepstrum for mation, the convolution of signals to additions of the respective cepstras. Both methods were previously performed on polyacrylic glass, and the thickness d was also measured with the caliper and the transit time method. The calculated thicknesses can be seen in table 8.

Method	d [mm]
Calliper	$9,9600 \pm 0,0050$
FFT	$9,4 \pm 1,3$
Cepstrum	$11,2 \pm 1,9$

Table 10: Comparison of different methods of length measurement

The spectrum (see Fig.5 above) shows many aquidistant peaks for the frequency of the reflected signal, and their overtones are recognizable. In the cepstrum (see Fig.5 below), fewer and broader peaks are recognizable. This can now also be used to determine the thickness of the polyacrylic glass.

7 Conclusion

In this experiment, the speed of sound in different materials was investigated using a GS200 ultrasonic echoscope. The speeds of sound were determined in polyacrylic, cooper, aluminum and water. The value for water is $v_w = (1,4507 \pm 0,0068)\frac{km}{s}$. The speeds of sound in cooper and aluminum can be found in table 6. The propagation speed of transverse and longitudinal waves was determined. A frequency dependence was investigated when the speed of sound in polyacrylic. The sound velocities are shown in table 4. A dispersion relationship could not be confirmed. The damping of a large polyacrylic body was determined at various frequencies. The results are shown in table 7. The damping constant was found to increase with increasing frequency. However, with increasing frequency, the resolution also increases. To measure a test block, as shown in Figure 3, a frequency of 2 MHz was chosen. This frequency has sufficient resolution but is not absorbed as quickly as 4 MHz. The measurement taken with ultrasound and a caliper agree due to the error. The thickness of a polyacrylic disc was determined using other spectral methods such as FFT and the cepstrumof the GS200 ultrasonic echoscope. The determined thicknesses are consistent with those measured using callipers.

Sources

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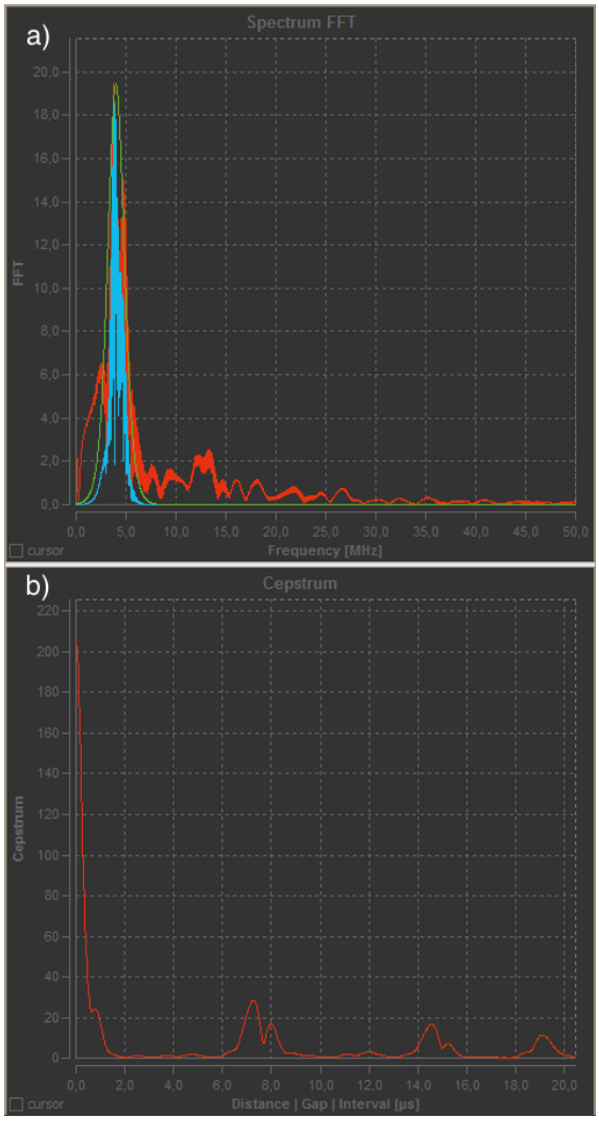

Figure 5: Spectral analysis of a polyacrylic block approximately 10 mm thick. a) Spectrum Fast Fourier Transform (FFT) of the measured signal, showing the dominant frequency components and their amplitudes. b) Cepstrum plot of the signal, identifying periodic structures and reflection distances within the material, visible as peaks at specific intervals.

8 Attachment

Differential equations are used to mathematically represent mechanical waves. An example of such a differential equation can be found in 4.

$$\rho * \frac{\partial^2 u_i}{\partial t^2} = \sum_{jkl} c_{ijkl} \frac{\partial^2 u_l}{\partial x_j \partial x_k} \tag{4}$$

The density of the medium through which the wave propagates is denoted by ρ, and u_i represents the displacement in the x_i direction. The elastic tensor c_{ijkl}, which characterizes the linear relationship between stress and strain in a given material, contains the elastic stiffness constants. For crystals with cubic symmetry, such as copper and aluminum, this fourth rank tensor reduces to three independent constants: c_{11}, c_{12}, and c_{44}. The constant c_{11} describes the elastic response under uniaxial deformation along one of the principal crystallographic axes and effectively corresponds to the compressional stiffness in these directions. The term c_{12} quantifies the coupling between the orthogonal deformation directions, determining how a strain along one axis induces stresses along the perpendicular axes. Finally, c_{44} represents the shear modulus associated with shear distortions in planes perpendicular to the cubic axes. Together, these three material parameters fully describe the linear elastic behavior of cubic crystals.

$$\begin{pmatrix} c_{11} & c_{12} & c_{12} & 0 & 0 & 0 \\ c_{12} & c_{11} & c_{12} & 0 & 0 & 0 \\ c_{12} & c_{12} & c_{11} & 0 & 0 & 0 \\ 0 & 0 & 0 & c_{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & c_{44} & 0 \\ 0 & 0 & 0 & 0 & 0 & c_{44} \end{pmatrix}$$

By solving equation 4, one obtains the three wave velocities v_1, v_2, and v_3 as a function of the propagation angle Φ in the x-y plane. If the direction of propagation is aligned with the principal axes ($\Phi = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$), the wave velocities can be reduced to the following equations.

$$v_1 = \sqrt{\frac{c_{11}}{\rho}}, v_2 = \sqrt{\frac{c_{44}}{\rho}} = v_3$$

The velocity v_1 is determined solely by the stiffness constant c_{11}, which represents a measure

of the longitudinal stiffness along the principal axis. This is a characteristic of a longitudinal wave. In contrast, velocities v_2 and v_3 are only affected by the shear stiffness c_{44} and therefore correspond to transverse modes. Here, mv_3 is independent of angle and was derived from the displacement component in the z-direction, which identifies it as an out of plane transverse wave. Thus, v_2 represents the transverse wave within the plane. Since the elasticity tensor is material dependent, these velocity values vary between different substances. Table 6 shows the relevant parameters for copper and aluminum. With these values, the wave velocities can be calculated and displayed for illustration in polar plans, as shown in figure 6 and 7. In aluminum, the velocities of the transverse waves are close to each other compared to copper. In copper, the transverse velocities drift further apart, and a stronger angular dependence can be observed. A measure of the anisotropy, and thus also of the angular dependence, is the factor a.

$$a = 2\frac{c_{44}}{c_{11} - c_{12}} \tag{5}$$

Here, c_{44} refers to the shear stiffness, c_{12} quantifies how normal stresses are coupled between perpendicular axes, and c_{11} shows how the crystal responds under longitudinal loading in a main direction. The expression $(c_{11}c_{12})$ indicates how the elastic properties change depending on the orientation, so that the ratio a specifies the magnitude of the velocity changes with direction. A larger a implies a stronger anisotropic response; in the special case of a=1, a material behaves isotropically. Using Table 11 and Equation (5), we get:

$$a_{aluminum} = 1,22343, a_{cooper} = 3,20851$$

Both materials are anisotropic because their a is greater than one. Because of the significantly higher value of copper, a stronger directional dependence is also expected for copper compared to aluminum. This assumption is confirmed in Figures 6 and 7.

$$v_1 = \sqrt{\frac{2(c_{11} + c_{44}) + x}{4\rho}} \tag{6}$$

$$v_2 = \sqrt{\frac{2(c_{11} + c_{44}) - x}{4\rho}} \tag{7}$$

$$v_3 = \sqrt{\frac{c_{44}}{\rho}} \tag{8}$$

$$x = \sqrt{2(A + B + C)} \tag{9}$$

$$A = c_{11}^2 + c_{12}^2 - 2c_{11}c_{44} \tag{10}$$

$$B = 2c_{22}(c_{12} + c_{44}) \tag{11}$$

$$C = (c_{11} + c_{12})(c_{11} - c_{12} - 2c_{44})cos(4\Phi) \tag{12}$$

Constants	Aluminum	Cooper
$c_{11}[10^{11}\frac{N}{m^2}]$	1,068	1,684
$c_{12}[10^{11}\frac{N}{m^2}]$	0,607	1,214
$c_{44}[10^{11}\frac{N}{m^2}]$	0,282	0,754
$\rho[\frac{g}{cm^3}]$	2,6795	8,899

Table 11: Material parameters (elastic constants c_{11}, c_{12}, c_{44}, as well as density ρ) of aluminum and copper, which are used to determine the longitudinal and transverse sound velocities.

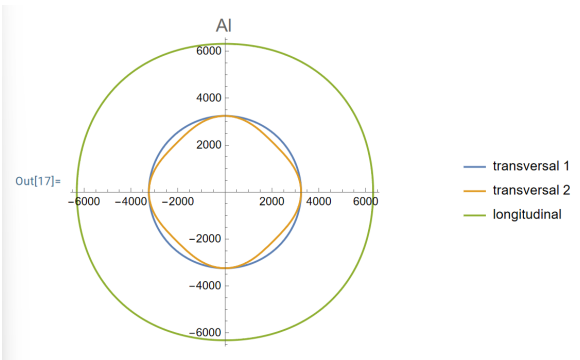

Figure 6: Theoretical wave velocity $[\frac{m}{s}]$ of transverse (on graph "transversal") and longitudinal waves in aluminum. Displays in a polar plot.

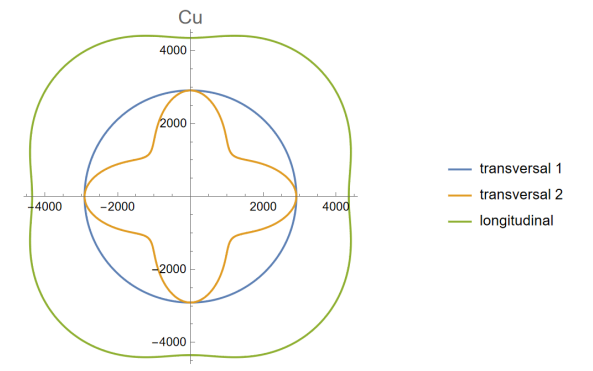

Figure 7: Theoretical wave velocity $[\frac{m}{s}]$ of transverse (on graph "transversal") and longitudinal waves in cooper. Displays in a polar plot.